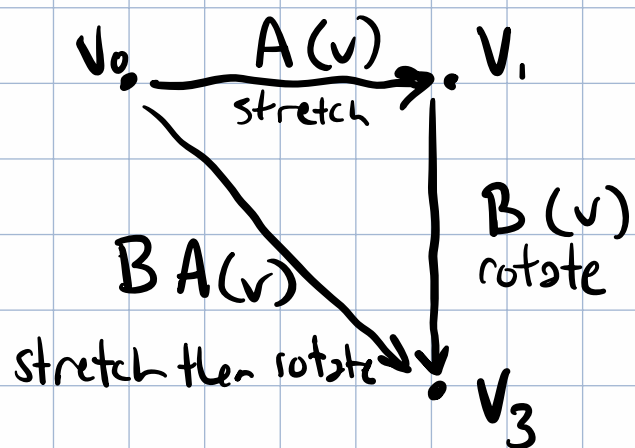


1) Given A stretches a vector, B rotates a vector by $\pi/2$, describe BA



$$BA(v) = (B \circ A)(v)$$

"B after A"

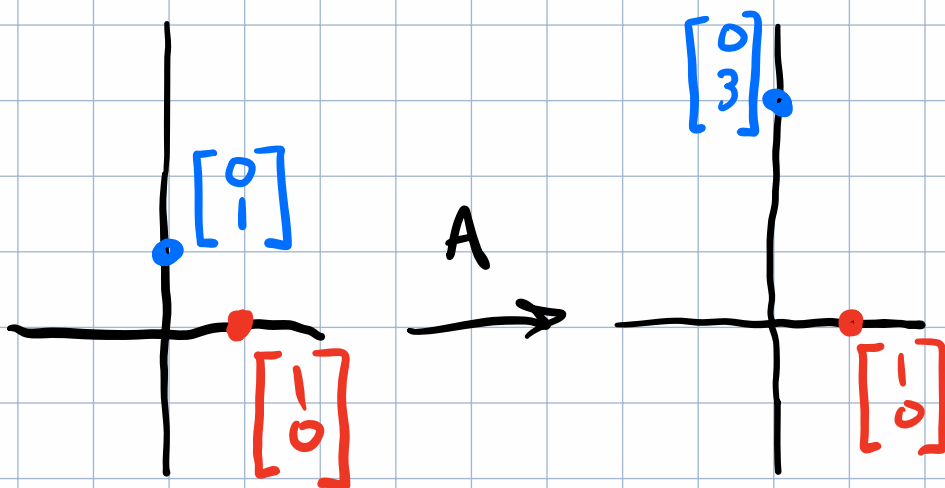
Matrix multiplication is isomorphic to function composition

objects are vectors

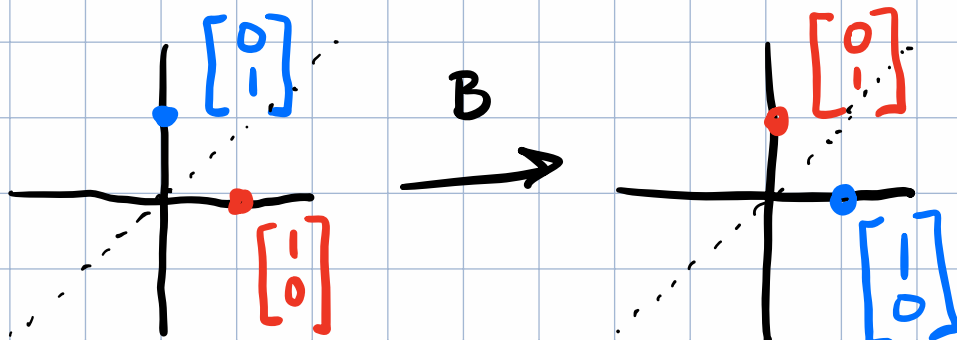
morphisms are matrices

identity morphism for $\mathbb{R}^2$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

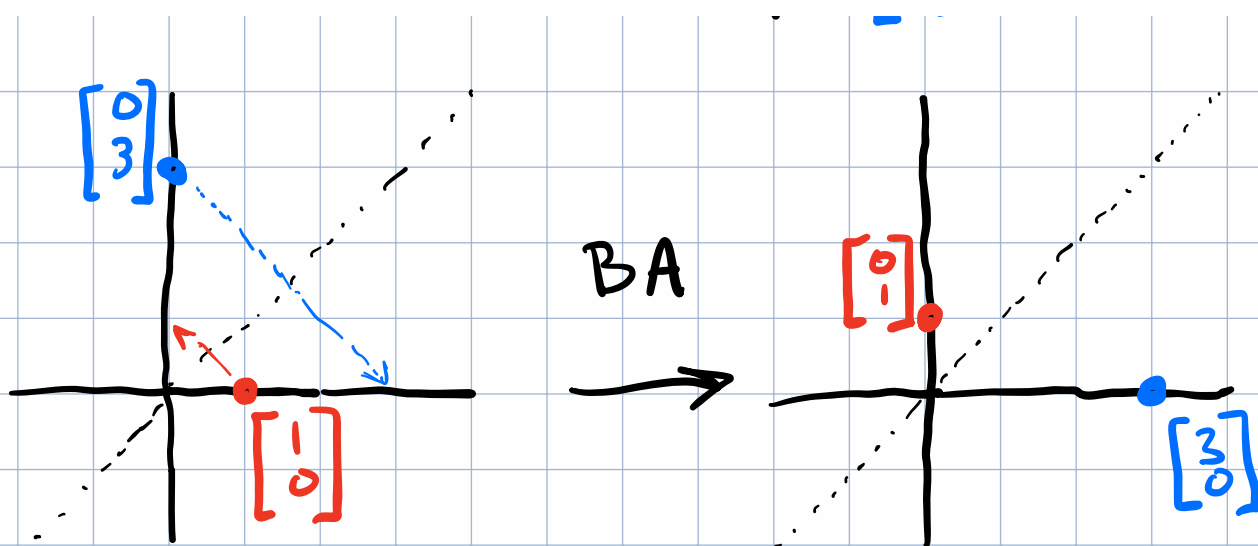
2) $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = A$ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = B$ without matrix multiplication, find BA



stretch y by 3



reflect along $y=x$



$$\therefore BA = \begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix}$$

- 3) It is enough to know where a matrix (linear transformation) sends a basis vector to know what it does to any vector because the linear transformation transforms the vector space, and the basis vectors are an easy representation of that transformation. All vectors will be transformed in the same way.

This lets us go from the Category of matrices to the Category of linear transformations of vector spaces.